

Supplementary Material

Corrected Statements and Proofs for the Revised Long-Memory Limit Theorems

Companion to `revised_theorems.tex`; machine-checked items verified in the Lean 4 repository at closure commit 16715db

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This supplement gives, for each substantive correction identified in the audit of 19-AOS1825.pdf: the original claim, the error (with the counterexample where one exists), the corrected statement, and a proof or proof sketch. Section 3 then proves in full the honest-rate drift proposition underlying the revised main theorem. Full written chains are in `direct_proofs/01–24`; per-item machine status is in `paper_revision/theorem_inventory.md`. Status tags: [Lean] = completely formalized; [written] = written mathematical proof. Proofs marked “English rendering” are faithful translations of the corresponding Chinese sources in `direct_proofs/`; every item also carries a pointer *Full written chain* to the exact source files, which remain authoritative for any detail not reproduced here.

1 Corrections to the main theorems

1.1 Critical-regime variance (Theorem 3.3(ii), via S.3.2)

Original. In the critical regime $H(t) = \frac{3}{4}$ the limiting variance is expressed through the kernel center value. **Error.** The derivation of the diagonal contribution ε_4 in S.3.2 replaces $f^2(i/N)f^2(j/N)$ by $f^4(0)$ for $|i - j|$ small, which is not summable to $f^4(0)$; the correct diagonal sum tends to $\int f^2$, and the Hermite rank-2 factor $2!$ together with the squared denominator of g was mishandled. **Corrected.**

$$\text{Var} \longrightarrow \frac{9}{16} \int \omega_r(x)^2 dx.$$

Proof (English rendering of `direct_proofs/03`, §5–6). For unit-stride first increments of fractional Brownian motion with constant H ,

$$r_H(k) = \frac{1}{2}(|k+1|^{2H} + |k-1|^{2H} - 2|k|^{2H}), \quad r_H(0) = 1.$$

At $H = \frac{3}{4}$ a Taylor expansion gives $r_{3/4}(k) = \frac{3}{8}|k|^{-1/2} + O(|k|^{-5/2})$, so both long-memory spectral coefficients equal $a(\pm 1) = \frac{3}{8}$ and $c_a = 2(\frac{3}{8})^2 = \frac{9}{32}$. The log-square statistic is a weighted quadratic form in the standardized increments; diagonalizing it as $Q_N = \sum_j \lambda_{N,j}(Z_j^2 - 1)$ with $s_N^2 = \sum_j \lambda_{N,j}^2$, the operator bound $\max_j |\lambda_{N,j}| \leq \|B_N\|_{\text{op}} = O(N^{1/2})$ gives $\max_j |\lambda_{N,j}|/s_N = O((\log N)^{-1/2}) \rightarrow 0$, and the independent- χ^2 log-characteristic function

$$\log \mathbb{E} e^{itQ_N/s_N} = \sum_j \left\{ -itu_{N,j} - \frac{1}{2} \log(1 - 2itu_{N,j}) \right\} = -\frac{t^2}{2} + o(1)$$

(with remainder $\leq C_t \max_j |u_{N,j}| \sum_j u_{N,j}^2 \rightarrow 0$) yields the Gaussian limit by Lévy continuity — notably *without* invoking the short-memory Breuer–Major theorem, whose summability hypothesis fails at criticality. The limiting variance is $2c_a \int f^2 = \frac{9}{16} \int \omega^2$.

Counterexample to the printed formula. Take $d = q = p = 1$, $H \equiv \frac{3}{4}$, and the smooth nonnegative bump $K_0(z) = z^2 e^{-1/(1-z^2)}$ on $|z| < 1$ (zero otherwise), normalized to $K = K_0 / \int K_0$, with t interior and $b = n^{-1/2}$. The equivalent kernel $\omega = K / \int K$ then satisfies $\omega(0) = 0$ but $\int \omega^2 > 0$. By fBm stationarity and self-similarity the standardized local increment vectors carry exactly the covariance $r_{3/4}$; the actual weights differ from $N^{-1}\omega(z_i)$ by $N^{-1}e_{N,i}$ with $\sup |e_{N,i}| = O(N^{-1})$, and the covariance bound $|\text{Cov}(g(Y_i), g(Y_j))| \leq Cr(i-j)^2$ makes the weight-perturbation variance $O(N^{-2})$ after critical normalization. Hence the centered $\hat{G}$ converges under the paper’s $\sqrt{N/\log N}$ normalization to a normal law with variance $\frac{9}{16} \int \omega^2 > 0$, while the paper’s kernel-center-value formula returns 0: a counterexample to the *universal critical variance formula itself*, not merely to a step of its derivation.

[written]

Full written chain: `direct_proofs/03`, `direct_proofs/13` (Chinese sources).

1.2 Long-memory sign (Theorem 3.3(iii)–3.4)

Original. The inverted estimator converges to $+Q$. **Error.** The inversion $x \mapsto (c_\sigma - x)/(2 \log n)$ carries a negative slope, so the limit reflects: $-Q$. This is substantive: a legal nonnegative equivalent-kernel instance has third moments of Q and $-Q$ of opposite sign, so the two laws differ. **Corrected.** $2S_n^\psi \log n (\hat{H}_n - h) \Rightarrow -Q$ (Theorem 1 of the companion note). **Proof (direct_proofs/14, §7; the full statement is machine-checked).** The calibration $x \mapsto (c_\sigma - x)/(2 \log n)$ is decreasing, so the expectation-centered limit of $\hat{G}$ transports to the *reflected* limit $-Q$; the affine inversion argument at $a_n = N^\psi$ carries the deterministic center to the true expectation center using $V_n = O(N^{-2\psi})$.

The sign is substantive. Take $p = 2$ and a nonnegative, symmetric, nonzero kernel K ; the local-linear equivalent kernel $\omega = K / \int K$ is then nonnegative, and for any admissible H with $H(t) > \frac{3}{4}$ (constant H suffices) the limiting quadratic-form operator $\mathcal{A}$ is nonnegative and nonzero: using the representation $\frac{1}{2}e^{-a|x-y|} = a \int_{-\infty}^{\min(x,y)} e^{-a(x-u)} e^{-a(y-u)} du$, the ω -weighted quadratic integral is an integral of nonnegative squares, and if it vanished then $F(u) = \int_u^\infty \omega(x) e^{-a(x-u)} dx \equiv 0$ together with $F' = aF - \omega$ would force $\omega = 0$, contradicting $\int \omega = 1$ (all exchanges are justified by the integrability of the kernel family). Hence $\text{tr}(\mathcal{A}^3) > 0$ and

$$\mathbb{E}Q^3 = 8 \text{tr}(\mathcal{A}^3) > 0 \quad \text{while} \quad \mathbb{E}(-Q)^3 < 0,$$

so Q and $-Q$ are not identically distributed — on a submodel allowed by the paper’s assumptions, at the level of the actual statistic’s limit. The fourth cumulant $48 \sum_j \lambda_j^4 > 0$ ($\mathcal{A} \neq 0$) also shows Q is not Gaussian.

[Lean]

Full written chain: `direct_proofs/14`, `direct_proofs/13` (Chinese sources).

1.3 Optimal-bandwidth biased means (Corollary 3.5)

Original. Both $\hat{G}$ and $\hat{H}$ limits carry mean $R(t)$. **Error.** The two means have opposite signs. **Corrected.** With $b/b_0 \rightarrow 1$, $b_0 = (n^d \log^2 n)^{-1/(2p+d)}$:

$$\text{mean}(\hat{G}) = -2R(t), \quad \text{mean}(\hat{H}) = 2R(t), \quad \text{variance} = \xi^2(t) \text{ in both;}$$

if $b/b_0 \rightarrow c$, multiply the means by c^p and the variances by c^{-d} . R may take either sign; $R = 0$ requires the o -term from the bias expansion, not an equivalence with 0. **Proof sketch.** The smoothing bias enters $\mathbb{E}\hat{G}$ with the sign fixed by the second-order expansion; the inversion flips it.
[written]; conditional Lean chain connected

Full written chain: `direct_proofs/13`, `direct_proofs/11`, `direct_proofs/01` (Chinese sources).

1.4 Growing-lag remainder (Theorem 8.2(iv))

Original. A stronger remainder is claimed. **Corrected.** For effective interior differences with $b \log n \rightarrow 0$,

$$\text{error} = O\left(k^{-2} + \frac{k \log n}{n} + \left(\frac{k}{n}\right)^\psi\right) \quad (q = 1), \quad O\left(k^{-2} + \frac{k \log n}{n}\right) \quad (q = 2),$$

and the finite-lag error must be retained. **Reason.** Cross terms between lag growth, mesh resolution and the Hölder exponent cannot be dropped.

[written]; interior pieces Lean

Full written chain: `direct_proofs/12`, `direct_proofs/18` (Chinese sources).

1.5 Local-polynomial bias (Lemma 8.4)

Original. An integral-region/quadrature argument. **Error.** The quadrature over the integration region is invalid. **Corrected.** Define $M_{kl} = \sum_i K_i A_{ik} A_{il}$ and $w_i = K_i \sum_l (M^{-1})_{0l} A_{il}$. **Proof.** Invertibility of M gives $\sum_i w_i A_{ik} = \delta_{k0}$ exactly; hence for $F_i = \beta_0 + \sum_{k \geq 1} \beta_k A_{ik} + r_i$ with $|r_i| \leq B$ on the support of K ,

$$\left| \sum_i w_i F_i - \beta_0 \right| \leq \sum_i |w_i| B = O(b^p),$$

using the uniform L^1 weight bound. No $(nb)^{-\min(2,p)}$ term is needed.

[Lean](designs, weights, equivalent kernel, bias leading term)

Full written chain: `direct_proofs/02`, `direct_proofs/18` (Chinese sources).

1.6 Vector Z limits (Lemma 8.5)

Original. Limits stated for Z itself; dimension a taken as d . **Corrections.** State the limits for $Z - \mathbb{E}Z$; keeping Z requires separately proving $\mathbb{E}Z \rightarrow 0$ after normalization — the bound $\mathbb{E}Z = O((nb)^{d/2}\rho)$ alone does not suffice. The dimension is the polynomial basis dimension S , not d ; kernel conditions added.

expectation-bias leading terms [Lean]; three Z-limits [written]

Full written chain: `direct_proofs/03`, `direct_proofs/13`, `direct_proofs/14` (Chinese sources).

2 Corrections to the supplementary lemmas

2.1 S.1.1 (Fourier adjacent differences)

Error. As a complex Lebesgue integral the claim fails: near 0 the imaginary part behaves like $-ikx/|x|^{2+\delta}$, which is not absolutely integrable. **Correction.** Replace $1 - e^{ikx}$ by $1 - \cos(kx)$, or define the principal value explicitly; the scaling and mean-value steps then give the adjacent-difference bounds.

[written](repair); not formalized

Full written chain: `direct_proofs/06` (Chinese source).

2.2 S.1.2 (filtered Fourier integral)

Status. No counterexample found; a more direct route exists but is not formalized. **Route.** Use the cosine form and the second difference $(k+1)^{1+\delta} - 2k^{1+\delta} + (k-1)^{1+\delta}$ in place of the contour argument; $k=1$ is handled separately and $\delta=0$ returns 0; the real part must be retained in the contour derivation.

Full written chain: `direct_proofs/06` (Chinese source).

2.3 S.2.1 (g bounded away from zero)

Error. Insufficient hypotheses: at $h=0$ the quantity vanishes, and for $q=2$, $H \rightarrow 1$ no uniform positive lower bound exists. **Correction.** Require $h \neq 0$ and $H \in [\gamma, 1-\gamma]$, the latter following from [A1] rather than [A3]; positivity, continuity and compactness then give the uniform lower bound.

repaired 1-d version [Lean]

Full written chain: `direct_proofs/05` (Chinese source).

2.4 S.2.2 / S.3.1 (leading singular coefficient)

Error. The printed c_h misses a factor $\frac{1}{2}$. **Correction.** $c_h = h(2h-1)$, i.e. $g(H, u, h) = (c_h + o(1))|u|^{-\psi}$ with $\psi = 2 - 2H$ (frozen H ; all derivatives acting on the distance in the principal term, H -derivatives in the remainder). The $o(1)$ must not be strengthened to $O(b \log b)$.

instantiated and machine-verified inside the closure chain [Lean]

Full written chain: `direct_proofs/12`, `direct_proofs/10`, `direct_proofs/04` (Chinese sources).

2.5 S.2.3 (mixed derivatives, near-diagonal)

Error. The $q=1$ near-diagonal remainder $O(r \log r)$ admits a lawful linear- H counterexample. **Correction.** Add the smooth background term of relative size r^ψ : the corrected bound retains the $x^{H(x)+H(y)-q}$ -times-log-power background near 0 and normalizes on the actual grid.

[written]; grid normalization pieces [Lean]

Full written chain: `direct_proofs/18`, `direct_proofs/12`, `direct_proofs/04` (Chinese sources).

2.6 S.3.2 (weighted power sums / cycle products)

Corrected jointly with §1.1 above (center value $\int f^2$; $2!$; denominator). The long-memory side — identification of the cycle products with the spectrum of $B = K^{1/2} M_\omega K^{1/2}$ for all orders $k \geq 2$ — is machine-checked.

long-memory identification [Lean]

Full written chain: `direct_proofs/03`, `direct_proofs/13`, `direct_proofs/14` (Chinese sources); the critical side is proved in §1.1 above.

2.7 S.5.1 (spatial average variance of I'_1)

Error. The far-distance count misses a factor m : with $H \equiv 7/8$, $q = p = 1$, $b = n^{-1/2}$ one has $\text{Var}(I_1) \approx \log^2 n \cdot n^{-1/2}$ against the claimed $O(\log^2 n \cdot n^{-3/4})$. **Correction.** The count is $m^{2\psi-1} n^{-2\psi}$ times the lag sum; merging weights at $m \approx 1/b$ gives $\text{Var}(I_1) \leq C \log^2 n \cdot T(n)$, which the original bandwidth condition still absorbs into the target term — the main scale rates are preserved.

[written]; the $q=1$ scale input it feeds is discharged in Lean

2.8 S.5.2 (truncated component I'_2)

Repair. On the original truncation interval, use first- and second-derivative bounds for $\log v$ together with the projection error; control the bias through the pilot bias and MSE, and the squared error of the centered spatial average through Jensen — no independence between the channels is assumed, so no fictitious independence gain is claimed. For $q = 1$ this nonlinear term vanishes identically.

Full written chain: `direct_proofs/16`, `direct_proofs/08` (Chinese sources).

2.9 S.6.1 (irregular-grid covariance)

Error. As printed (unnormalized covariance) the claim fails already for Brownian motion. **Correction.** Either estimate $\text{Cov}(W_n(t), W_n(s))$ with $W_n(t) = \sigma^{-1} n^{H(t)} \Delta^q X(t)$, or restore $\sigma^2 n^{-H(t)-H(s)}$ on both sides of the original.

scalar counterexample [Lean]; full instantiation open

Full written chain: `direct_proofs/05` (Chinese source).

2.10 S.7.2 (dimension scope)

No counterexample; scope remark: the $d = 2, 3$ “minimax optimality” assertion has no corresponding lower bound in the manuscript itself, so it cannot be certified even after repairing the upper-bound route.

[written](route only)

Full written chain: `direct_proofs/02` (Chinese source).

3 Proof of the honest-rate drift proposition

This section proves Proposition 2 of the companion note in full. Work under (R1)–(R6) of the companion note: $\delta_n = n^{-h}$, $S_n = n^{1-h}$, $\psi = 2 - 2h \in (0, \frac{1}{2})$, $c_\sigma = \mathbb{E} \log Z^2$, $\widehat{G}_n$ the unit-weight log-square statistic, $\widetilde{H}_n = (c_\sigma - \widehat{G}_n)/(2 \log n)$, $\widehat{H}_n = T(\widetilde{H}_n)$ clipped to $[0, 1]$.

Proposition 1. $2S_n^\psi \log n |\mathbb{E} \widehat{H}_n - h| \rightarrow 0$, with the three-term bound of the companion note; the second term vanishes exactly under the b -band $(1 - h)^2 < 1 - b$, the other two unconditionally.

Proof. Step 1 (exact expectation identity). For each active row j the coefficient-feature combination satisfies, by the stride identity, the norm decomposition

$$\log \|\text{combo}_j\|^2 = -2H_j \log n + \log \|V_j\|^2,$$

where $H_j = f(s_j)$ is the left-endpoint Hölder value and V_j the normalized varying increment. The Gaussian expectation formula gives

$$\mathbb{E} \widehat{G}_n = \sum_j w_j (-2H_j \log n + \log \|V_j\|^2) + c_\sigma, \quad \sum_j w_j = 1,$$

whence, with c_σ pinned,

$$\mathbb{E} \widetilde{H}_n - h = -\left(\sum_j w_j H_j - h\right) - \frac{1}{2 \log n} \sum_j w_j \log \|V_j\|^2. \quad (1)$$

Step 2 (deterministic side). (i) *Weight contraction.* On the active window $|s_j - t| < \delta_n$, so Hölder regularity gives $|H_j - h| \leq D(1 + M)\delta_n$, and $|\sum_j w_j H_j - h| \leq \sum_j |w_j| |H_j - h| \leq C_{\text{st}} \delta_n^p$ with $p \geq 1$ (the sharper exponent uses the class regularity; $p = 1$ suffices). (ii) *Second-order increment log rate.* Write $V_j = F_k(s_j + \ell)$ -type normalized varying increment with $|k - h| \leq K\ell$, $\ell = 1/n$, $K = D(1 + M)$. The exact kernel identity

$$2\langle \text{frozen, mismatch} \rangle = \Lambda \Phi(m) - \Phi(h), \quad \Phi(q) = (s + \ell)^{2q} - s^{2q} + \ell^{2q}, \quad m = \frac{h+k}{2},$$

with $\Lambda - 1 = -\frac{1}{2} \|F_k(1) - F_h(1)\|^2 = O((k - h)^2)$ (uniform parameter Lipschitz; no smoothness of the normalizing constant needed) and the mixed-power Lipschitz bound $|\Phi(m) - \Phi(h)| \leq C|k - h|\ell$ (valid since $2a > 1$ absorbs the logarithmic sensitivity) yields

$$|\|V_j\|^2 - 1| \leq C(1 + K + K^2 + K^3) \ell^{2-2b} = C n^{-(2-2b)},$$

hence $|\log \|V_j\|^2| \leq 2C n^{-(2-2b)} =: C_{\log} n^{-(2-2b)}$ once small, and $|\sum_j w_j \log \|V_j\|^2| \leq C_w C_{\log} n^{-(2-2b)}$ with $C_w = \sum_j |w_j|$ uniformly bounded.

Step 3 (random side; tail-based transfer). Let $m = \mathbb{E}\tilde{H}_n$. Since $\mathbb{E}\tilde{H}_n \rightarrow h \in (3/4, 1)$ (companion note, center-band discharge), eventually $m \in [d', 1 - d']$ with $d' = (1 - h)/2$. On $\{\tilde{H}_n \notin [0, 1]\}$ we have $|\tilde{H}_n - m| \geq d'$. The unit-weight variance bound and the normalized energy give

$$V_n := \text{Var}(\tilde{H}_n) = \frac{\text{Var}(\hat{G}_n)}{4 \log^2 n} \leq \frac{4 g_{\text{LSV}} U^2 C_2}{4 \log^2 n} S_n^{-2\psi} = O\left(\frac{n^{-4(1-h)^2}}{\log^2 n}\right),$$

and Chebyshev gives $\varepsilon_n := \mathbb{P}(|\tilde{H}_n - m| \geq d') \leq V_n/d'^2$. Since $|T(x) - x| \leq |x - m|$ off $[0, 1]$ and $T(x) = x$ on $[0, 1]$,

$$|\mathbb{E}\hat{H}_n - \mathbb{E}\tilde{H}_n| \leq \mathbb{E}[|\tilde{H}_n - m| \mathbf{1}_{\tilde{H}_n \notin [0, 1]}] \leq V_n + \varepsilon_n \leq V_n(1 + d'^{-2}).$$

Step 4 (exponents). Combining $\mathbb{E}\hat{H}_n - h = [\mathbb{E}\tilde{H}_n - h] + [\mathbb{E}\hat{H}_n - \mathbb{E}\tilde{H}_n]$ with Steps 2–3,

$$2S_n^\psi \log n |\mathbb{E}\hat{H}_n - h| \leq \underbrace{C_1 \frac{\log n}{n^{h-2(1-h)^2}}}_{\text{smoothing}} + \underbrace{C_2 n^{-(2-2b)-2(1-h)^2}}_{\text{mismatch}} + \underbrace{C_3 \frac{1}{n^{2(1-h)^2} \log n}}_{\text{clipping}}.$$

Indeed: the smoothing term is $2S_n^\psi \log n C_{\text{st}} \delta_n^p$ with $S_n^\psi = n^{2(1-h)^2}$ and $\delta_n^p = n^{-hp}$, and $2(1-h)^2 - hp \leq 2(1-h)^2 - h < 0$ since $2(1-h)^2 < \frac{1}{8} < \frac{3}{4} < h$; the mismatch term is $S_n^\psi C_w C_{\log} n^{-(2-2b)}$ with exponent $2(1-h)^2 - (2-2b)$, negative exactly when $(1-h)^2 < 1-b$, i.e. (R4); the clipping term is $2S_n^\psi \log n V_n(1 + d'^{-2}) = O(n^{2(1-h)^2} \log n \cdot n^{-4(1-h)^2} / \log^2 n)$, unconditionally vanishing. $\square$

Remark 2 (Sharpness). The mismatch rate is attained: the expansion of $\log \|V_j\|^2$ has leading coefficient $f'(s)s^{2h-1}[2hD'/D + 2h \log s + 1]$ evaluated at the window, generically nonzero, so no per-row expansion order reaches the rate n^{-h} and the manuscript's original bias envelope at that rate fails for non-constant profiles. The window refutations are machine-checked; the lower-bound statement is derived by hand and registered.

4 Formalization cross-reference

Every [Lean] tag above is a theorem in the repository, axiom-checked within `{propext, Classical.choice, Quot.sound}`, zero sorry; the entry points are `actualQ1_knownScaleH_fullChain_honestRate` (main theorem), `honestRate_drift_tendsto_zero` (Proposition 1), and the audit theorems listed in the companion note's provenance table. Automated re-verification: `verification/final_acceptance.py`.